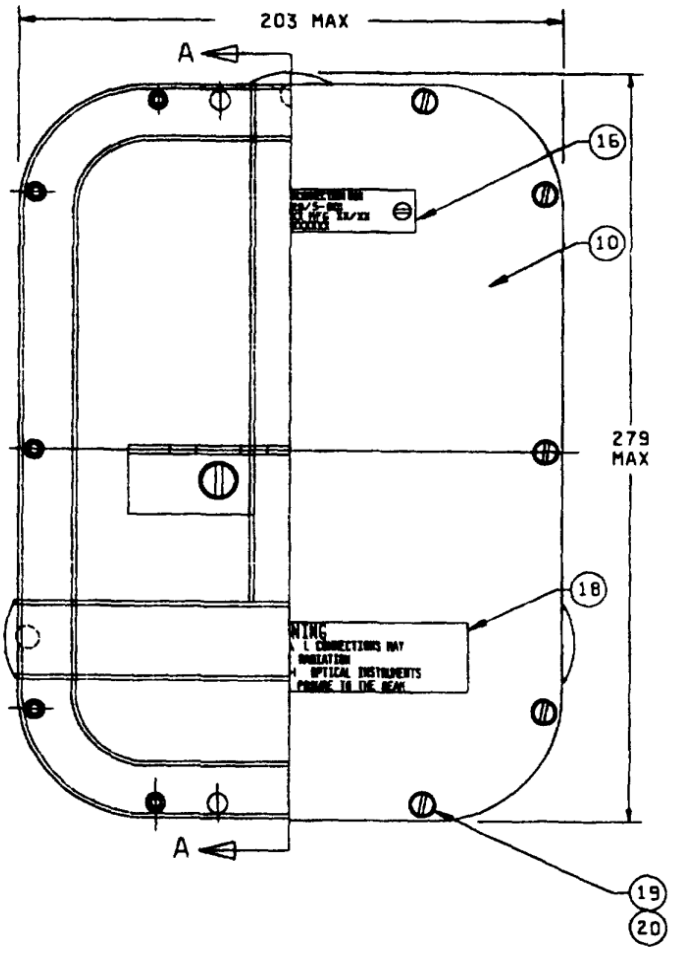
METRIC
MIL-DTL-24728/5
w/AMENDMENT 1
24 March 2023
SUPERSEDING
MIL-I-24728/5
1 June 1995

DETAIL SPECIFICATION SHEET

INTERCONNECTION BOX, FIBER OPTIC, SUBMERSIBLE, 152.4 X 228.6 MM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24728.



ABOVE COVER SHOWN PARTIALLY REMOVED

FIGURE 1. Enclosure assembly, M24728/5-001.

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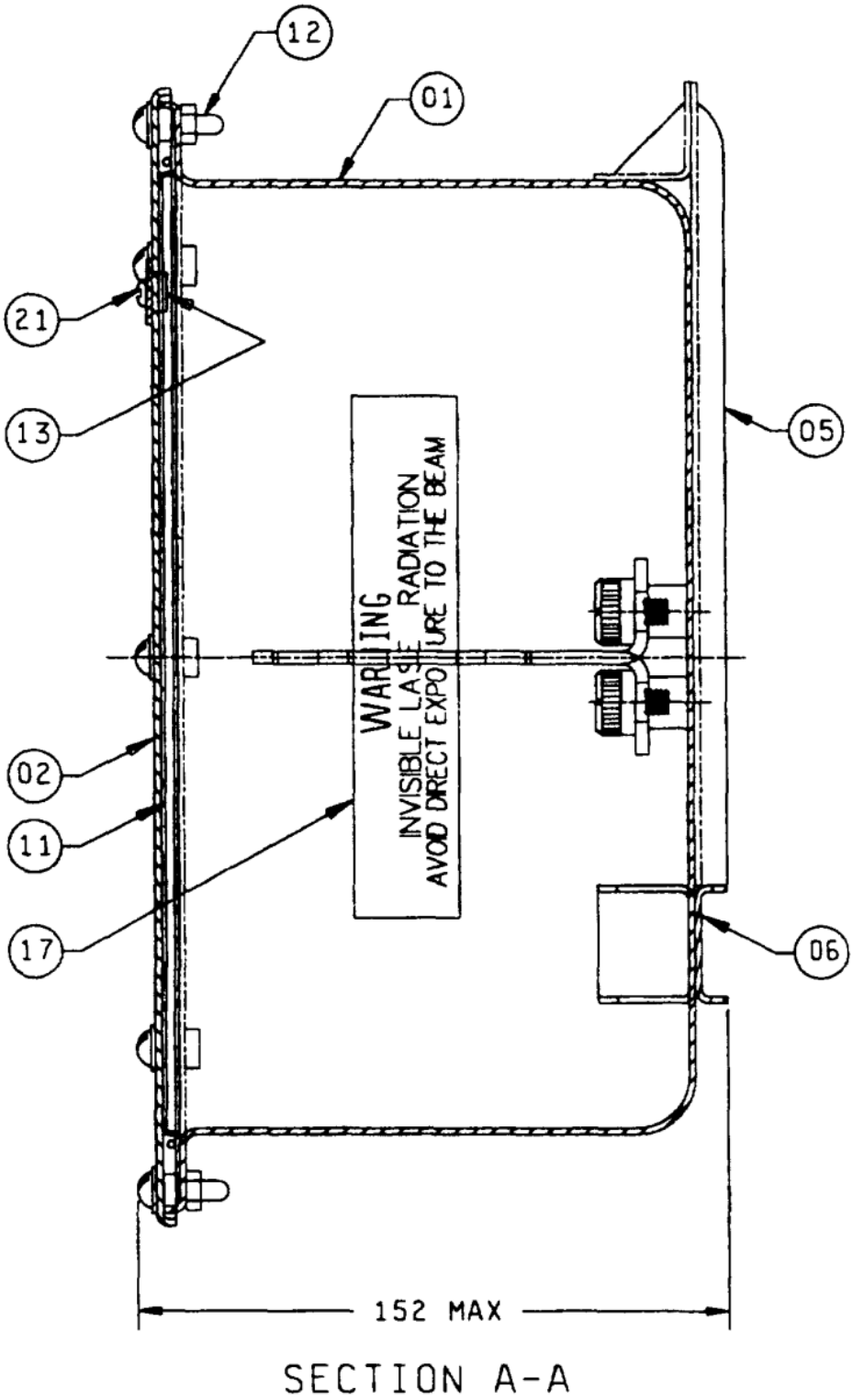
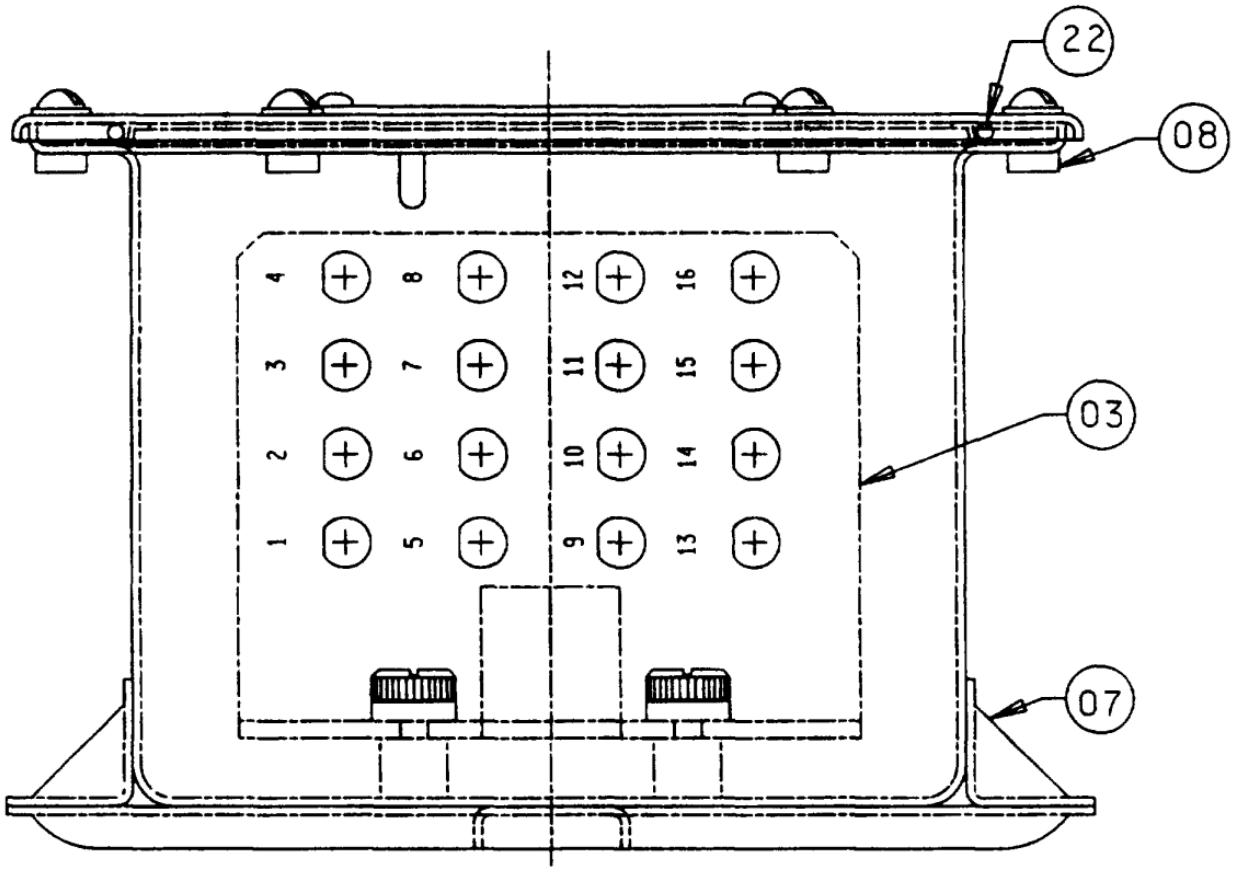


FIGURE 1. Enclosure assembly, M24728/5-001 – Continued.

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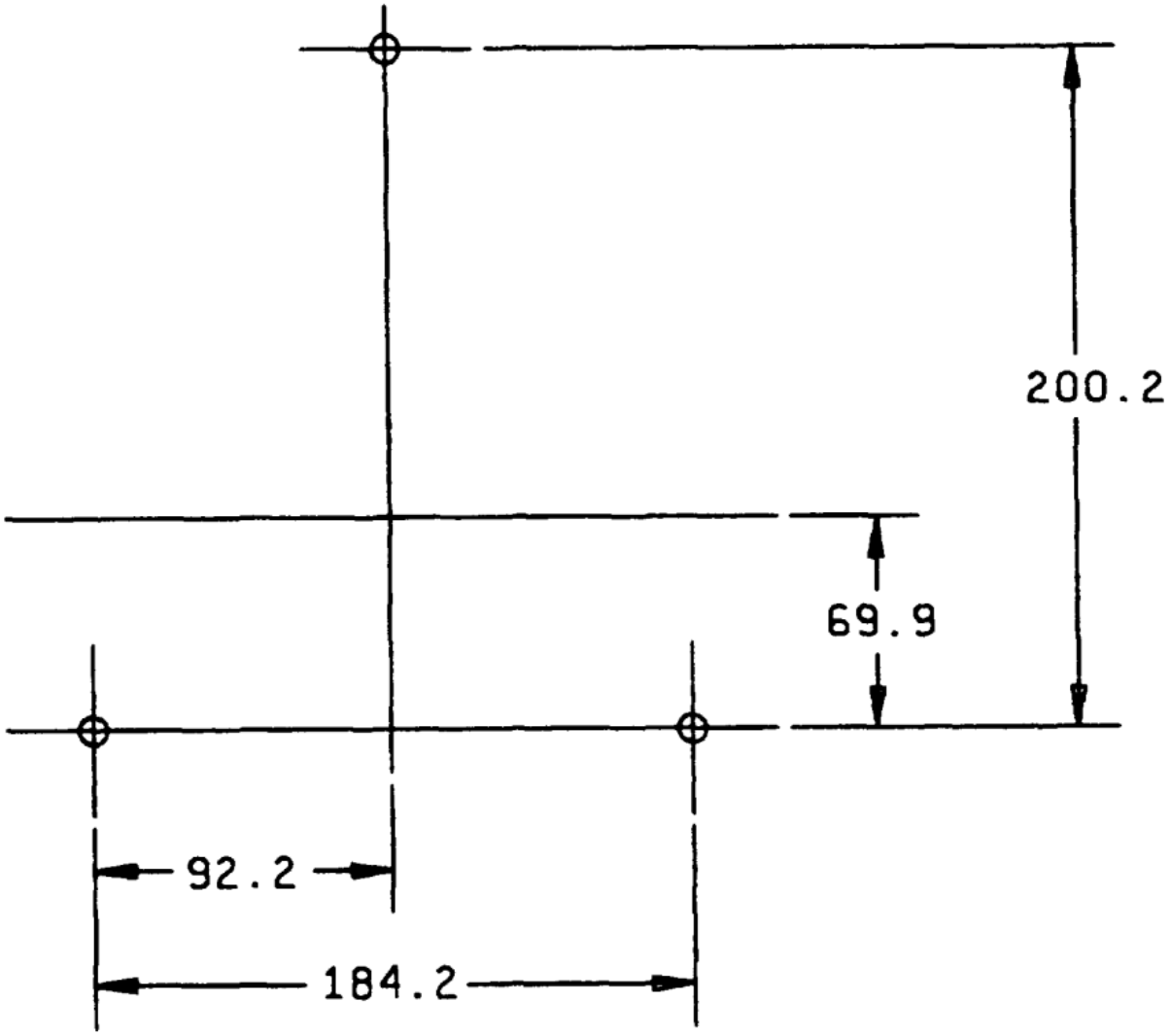
Millimeters	Inches
152	6.0
203	8.0
279	11.0

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for information only.
3. DELETED.
4. For identification of subcomponent parts, see table I.

FIGURE 1. Enclosure assembly, M24728/5-001 – Continued.

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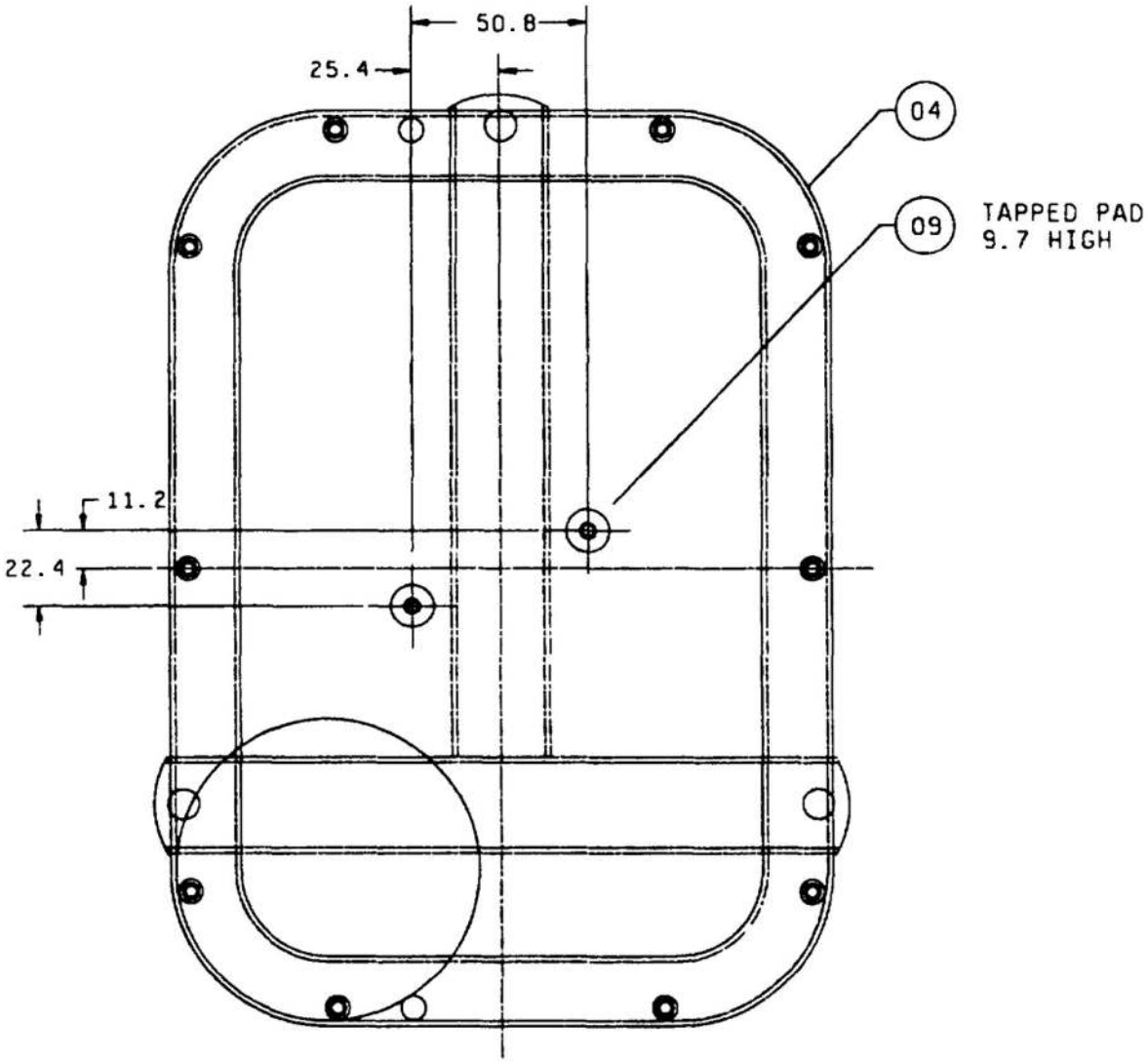
Millimeters	Inches
9.53	0.375
69.9	2.75
92.2	3.63
184.2	7.25
200.2	7.88

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for information only.
3. Dimensions are controlled by welding or brazing.
4. Mounting holes shall accommodate M9.53 x 1.59 (0.375-16) bolts.

FIGURE 2. Box mounting dimensions.

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Millimeters	Inches
6.4	0.25
9.7	0.38
11.2	0.44
22.4	0.88
25.4	1.00
50.8	2.00

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for information only.
3. DELETED.
4. Pad to have M6.35 x 1.27 (0.25-20) tap.

FIGURE 3. Location of patch panel mounting pads.

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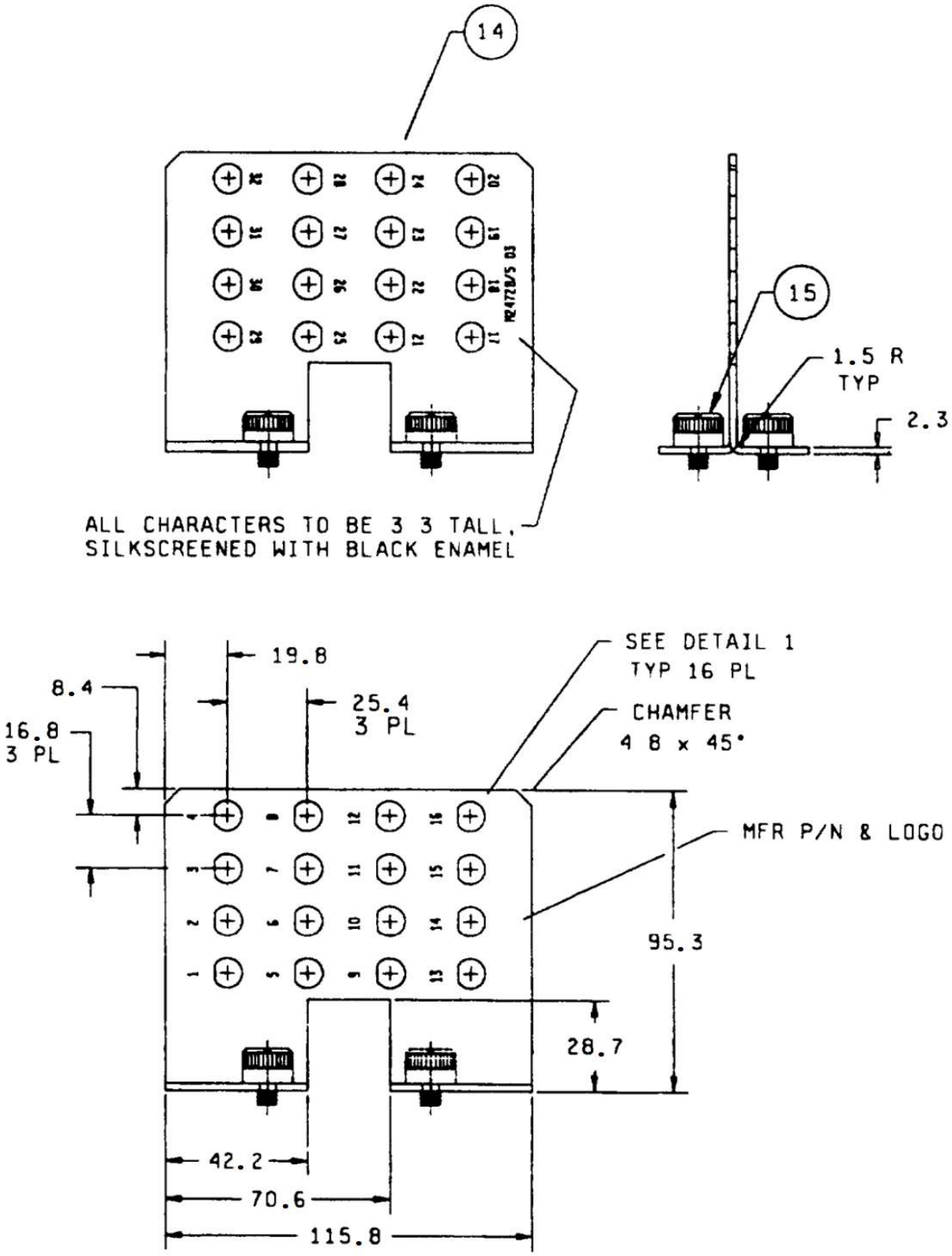
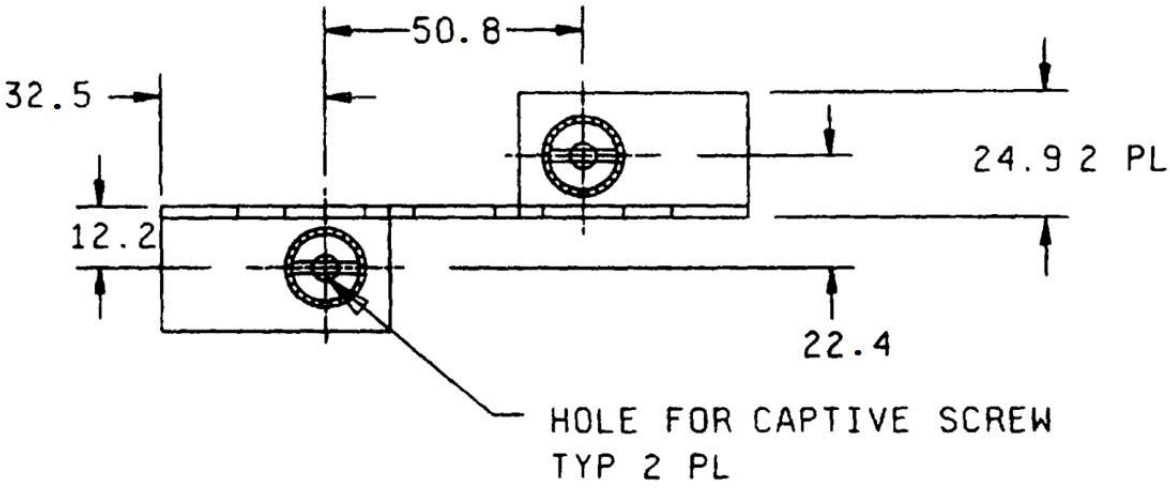
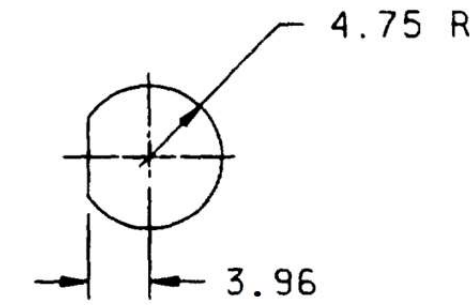


FIGURE 4. Optical connector patch panel assembly.

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DETAIL 1

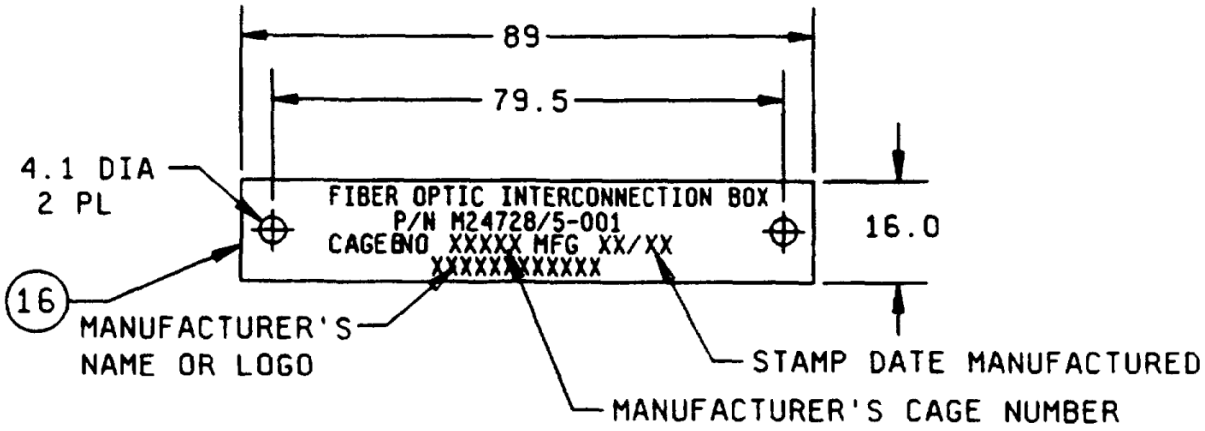
Millimeters	Inches	Millimeters	Inches	Millimeters	Inches
1.5	0.06	12.2	0.48	32.5	1.28
2.3	0.09	16.8	0.66	50.8	2.00
3.3	0.13	19.8	0.78	70.6	2.78
3.96	0.156	22.4	0.88	95.3	3.75
4.75	0.187	24.9	0.98	115.8	4.56
4.8	0.19	25.4	1.00		
8.4	0.33	28.7	1.13		

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for information only.
3. Tolerance is ±0.3 mm (0.01 inch) for two-place decimals.

FIGURE 4. Optical connector patch panel assembly – Continued.

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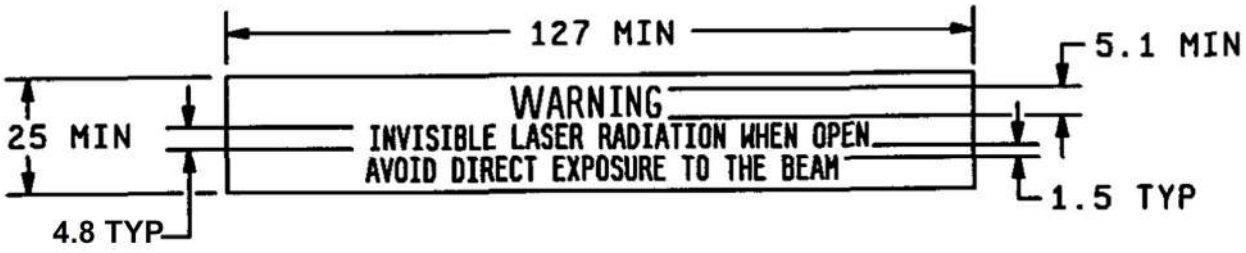
Millimeters	Inches
4.1	0.16
16.0	0.63
79.5	3.13
89.0	3.5

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for information only.
3. DELETED.

FIGURE 5. Identification plate.

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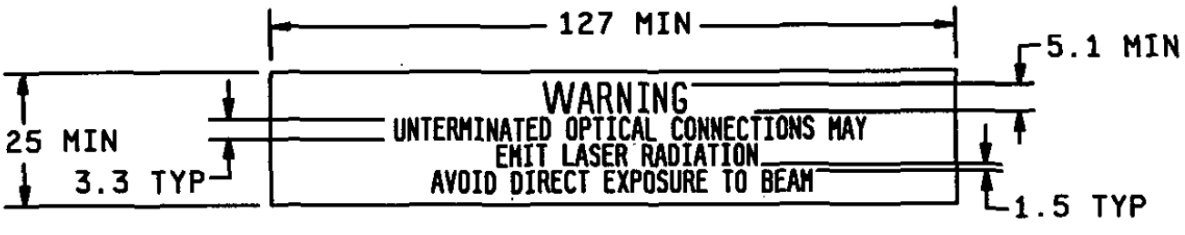
Millimeters	Inches
1.5	0.06
4.8	0.19
5.1	0.20
25	1.0
127	5.0

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for information only.
3. DELETED.

FIGURE 6. Inside warning label.

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Millimeters	Inches
1.5	0.06
3.3	0.13
5.1	0.20
25	1.0
127	5.0

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for information only.
3. DELETED.

FIGURE 7. Outside warning label.

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TABLE I. Parts list.

Subcomponent identification numbers, circled (see figures 1 through 7)	Part name	PIN M24728/5 dash number		Subassembly number			Remarks and reference figures
		-001	-002	01	02	03	
01	Box assembly	1		1			1 and 3
02	Cover assembly	1			1		1
03	Patch panel assembly	1				1	1 and 4
04 1/, 2/	152.4 x 228.6 box			1			1
05 1/, 3/	Mounting leg			1			1
06 1/, 3/	Mounting leg			1			1
07 1/, 4/	Gusset			3			1
08 1/, 5/	Weld nut			10			1
09 1/	Panel mounting pad			2			3
10 1/, 2/	152.4 x 228.6 cover				1		1
11 1/, 2/	Gasket retainer				1		1
12 1/, 2/	Dowel pin				2		1
13 1/	M3.5 x 0.79 (6-32) pad				2		1
14 1/	Patch panel		1			1	4
15 6/, 8/	M6.35 x 1.27 (0.25-20) captive screw		2			2	4
16 1/	Identification plate	1					5
17 7/	Warning label (inside)	1					6
18 7/	Warning label (outside)	1					7
19 1/, 2/	M2.5 x 1.06 (10-24) x 12.7 round head machine screw (RHMS)	10					1
20 1/	2.5 (#10) flat washer	10					1
21 1/	M3.5 x 0.79 (6-32) x 6.35 RHMS	2					1
22 2/	Gasket	1					1

NOTES:

1/ Brass alloy C26000 in accordance with ASTM B36/B36M.

2/ In accordance with MIL-E-24142/3.

3/ In accordance with MIL-E-24142, mounting channel figure.

4/ In accordance with MIL-E-24142, mounting bracket figure.

5/ In accordance with MIL-E-24142, weld nut figure.

6/ Corrosion-resistant steel (CRES).

7/ In accordance with MIL-DTL-15024, type G, adhesive-backed, metal foil.

8/ In accordance with MIL-DTL-24728/6.

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REQUIREMENTS:

Design and construction. See figures 1 through 7 and table I.

Interconnection box. The interconnection box shall be in accordance with MIL-E-24142.

Degree of enclosure. Submersible.

Enclosure assemblies construction. Enclosure assemblies shall be class 1A conversion coated in accordance with MIL-DTL-5541 before being epoxy powder coated.

Enclosure assemblies stress relief. Enclosure assemblies shall be stress relieved in accordance with MIL-E-24142.

Mass. M24728/5-001: 3.63 kilograms (8 pounds) maximum.

Interconnection box mounting. See figure 2.

Interconnect organization. Fiber optic connector adapters: The interconnection box shall organize up to 16 MIL-DTL-83522/17 connector adapters.

Cable retention. Not applicable.

Cable seal flexing. Not applicable.

Cable twist. Not applicable.

Compression resistance. Not applicable.

Impact. Not applicable.

Flammability. Not applicable.

Salt spray. Not applicable.

Flame spread. Not applicable.

Fungus test. The fungus test shall be conducted only on polymeric materials.

Operating and storage temperature range. See temperature range 1 of MIL-DTL-24728.

Part or identifying number (PIN). See table I.

M24728/5-001: Box and cover with patch panel assembly.

M24728/5-002: Patch panel assembly.

Qualification by similarity. Manufacturers who produce products for both MIL-DTL-24728/1 and this specification sheet and are qualified under MIL-DTL-24728/1, and pass the water pressure test, shock test, and size inspection specified herein, are qualified under this specification sheet.

Measurement system. The U.S. Government preferred system of measurement is the metric (SI) system. However, this item was originally designed using inch-pound units of measurement. In the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

AMENDMENT NOTATIONS: The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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CONCLUDING MATERIAL

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Army – CR
Navy – SH
Air Force – 85

Review activities:
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Navy – AS, CG, EC, MC, YD
Air Force – 19
DLA – CC

Civil agency:
NASA – NA

Preparing activity:
Navy – SH
(Project 6099-2022-009)

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